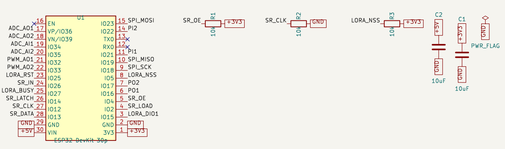
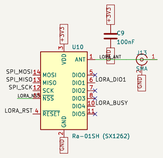
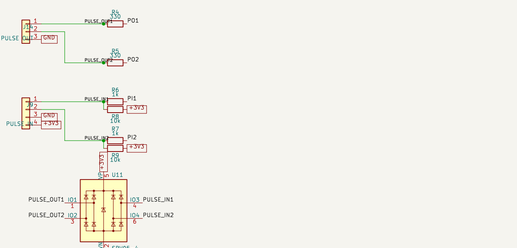
ESP32 DevKit 30-pin (USB powered: VIN = USB 5V, 3V3 from DevKit LDO)



LoRa SNIFFER – SX1262 module (WM1-Pro pins, except DIO1 on IO15)

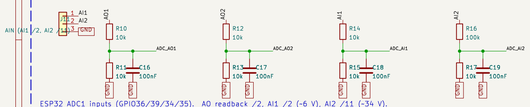


PULSE I/O – J14 out (LEDC/RMT), J9 in (PCNT)



PO1/PO2: LEDC/RMT pulse out via 330R. PI1/PI2: PCNT in via 1k, 10k pull-up (open-collector OK). U11 SRV05-4 clamps all four lines.

ESP32 ADC1 FRONT-ENDS – AO readback /2, AI1 /2, AI2 /11

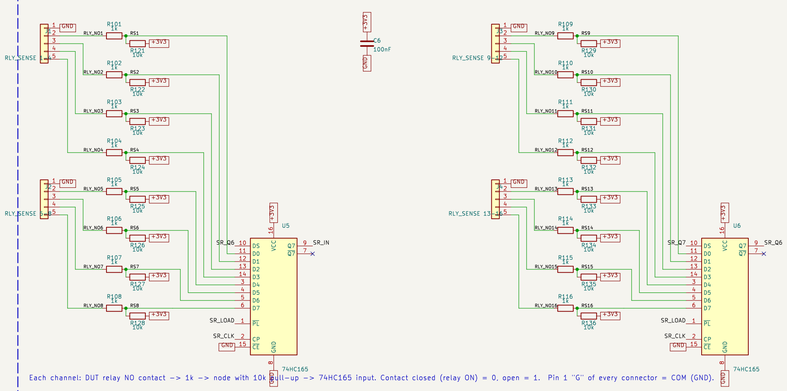


ESP32 ADC1 inputs (GPIO36/39/34/35). AO readback /2, AI1 /2 (÷6 V), AI2 /11 (÷34 V).

MECHANICAL

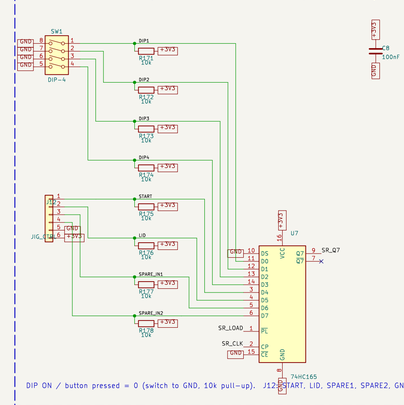


DUT RELAY SENSE – COM + NO, 4x JST-XH5 (pin1 = COM/GND). Relay ON -> NO closed -> reads 0



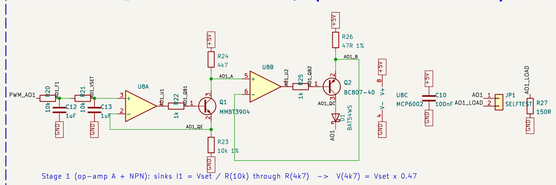
Each channel: DUT relay NO contact -> 1k -> node with 10k pull-up -> 74HC165 input. Contact closed (relay ON) = 0, open = 1. Pin 1 "G" of every connector = COM (GND).

JIG INPUTS – 4-way DIP, START, LID, SPARE1/2 on J12 (3rd 74HC165)



DIP ON / button pressed = 0 (switch to GND, 10k pull-up). J12: START, LID, SPARE1, SPARE2, GND, 3V3.

4-20 mA SOURCE #1 – Iout = Vset x 10 mA/V (0.4 V = 4 mA, 2.0 V = 20 mA), 5 V, load <= ~160R

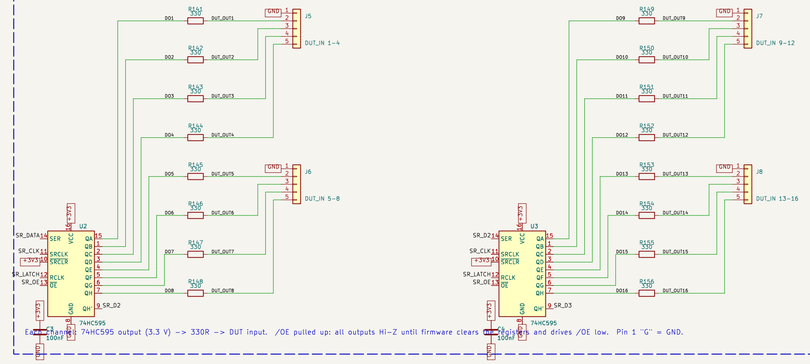


Stage 1 (op-amp A + NPN): sinks I1 = Vset / R(10k) through R(4k7) -> V(4k7) = Vset x 0.47

Stage 2 (op-amp B + PNP): forces the same drop across R(47R) -> Iout = Vset x 4k7 / (10k x 47R) = Vset x 10 mA/V

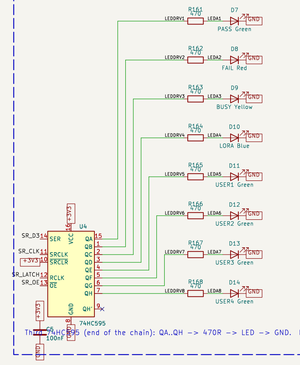
PWM 12-bit, 2-pole RC (10k/1uF x2) -> Vset. D1 blocks reverse current. JP1 + 150R = local self-test load (ADC readback).

DUT INPUT DRIVE – 74HC595 @3.3V, 4x JST-XH5 (pin1 = GND)



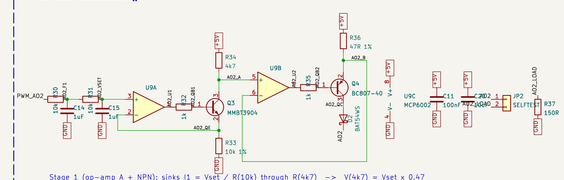
Each channel: 74HC595 output (3.3 V) -> 330R -> DUT input. /OE pulled up: all outputs Hi-Z until firmware clears it. Register and drives /OE low. Pin 1 "G" = GND.

JIG LEDs – 3rd 74HC595



Third 74HC595 (end of the chain): QA..QH -> 470R -> LED -> GND. PASS, FAIL, BUSY, LORA, USER1-4.

4-20 mA SOURCE #2



Stage 1 (op-amp A + NPN): sinks I1 = Vset / R(10k) through R(4k7) -> V(4k7) = Vset x 0.47

Stage 2 (op-amp B + PNP): forces the same drop across R(47R) -> Iout = Vset x 4k7 / (10k x 47R) = Vset x 10 mA/V

PWM 12-bit, 2-pole RC (10k/1uF x2) -> Vset. D1 blocks reverse current. JP1 + 150R = local self-test load (ADC readback).

2x 4-20mA (5V), 4x ESP32 ADC1, SX1262 LoRa sniffer, 4-way DIP, 8 LEDs. 100x100 mm 2-layer.
USB powered from ESP32 DevKit, 16 relay sense (COM+NO), 16 DUT outputs, 2+2 pulse.
Draft for review — generated by Claude

Sheet: /	
File: TestJig.kicad_sch	
Title: TestJig-ESP32 — DUT test jig (WM1 product range)	
Size: A1	Date: 2026-10-16
KiCad E.D.A. 9.0.7	**Rev: A**
	Id: 1/1